

Ex 1.5:-

"of EA rule (1.5.2) one $\rightarrow$ In his E's given matrix's another matrix obtained by performing

Definition:-

Note:-

different operations on Identity matrix. Now after that take a product of EA. Perform some operations on A for E. if $EA = A$ Then it means rule verified"

$$\frac{x \cdot 1}{x} = 1$$

So in terms of matrices $\therefore AA^{-1} = I$

$\therefore A^{-1}A = I \rightarrow$ Identity matrix. which is known as a matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

with 0, 1 with 1 in diagonal position.

For ex: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Solving equations with matrices:-

$$XA = B$$

$$\text{identity} \leftarrow XI = BA^{-1}$$

$$(BA)^{-1} \rightarrow A^{-1}B^{-1}$$

Note: $\rightarrow$ An matrix multiplied by its inverse gets us identity matrix

$\rightarrow$ only possible if 2 matrices have appropriate dimension

$\rightarrow$ Also matrix multiplication is not commutative.

If $\det(A) = 0$ then $A^{-1} = \frac{1}{0} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ "undefined" so it is called a singular matrix."

Inverse of 3x3 matrix:-

$$[A|I] \rightarrow [I|A]$$

Q). $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$

"If a row of 0's comes on left side not invertible."

Ans:- step 1:- Generate matrix of minors

"each entry is a particular determinant"

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -24 & -20 & -5 \\ -18 & -15 & -4 \\ 5 & 4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

"matrix of cofactors."

$$\begin{bmatrix} 1(0) - 24 \rightarrow -24 \\ \text{(etc)} \end{bmatrix}$$

step 2:- Generate matrix of cofactors

$$\begin{bmatrix} -24 & - & -5 \\ -18 & -15 & - \\ 5 & - & 1 \end{bmatrix}$$

$\rightarrow$ "corner and centre entries will remain they are."

$$\begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{bmatrix}$$

$\rightarrow$ "other four entries will have their signs inverted."

step 3:- Find the Adjugate / Adjoint

step 4 Divide Adjugate by Det

$$\begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ -5 & 4 & 1 \end{bmatrix}$$

$\rightarrow$ Diagonals remains same they don't change

$\rightarrow$ swap swap

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$$\begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ -5 & 4 & 1 \end{bmatrix} \div \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} \rightarrow 1$$

$$1 \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 & 4 \\ 5 & 0 \end{bmatrix} + 3 \begin{bmatrix} 0 & 1 \\ 5 & 6 \end{bmatrix} = -24 + 20 + 15 = 11$$

Ex 1.50-

- Q1) (a) Elementary
(b) Not elementary
(c) NOT
(d) NOT elementary
Q2) (a) Elementary
(b) Not elementary
(c) Not elementary
(d) No elementary
Q3) (a) Ans: $R_1 + \frac{1}{2}R_3$

Q4) (b)

Ans: Divide Row by $R_3/3$

Q1(c)

Ans: Interchange R_1 and R_4

(d)

Ans: $R_1 + \frac{1}{2}R_3$

Q11) (a) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

Ans: $\begin{bmatrix} 40 & 26 & -25 \\ 32 & 25 & -6 \\ -1 & 0 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 40 & -26 & -15 \\ -32 & 25 & 6 \\ -1 & 0 & 8 \end{bmatrix}$
 $\begin{bmatrix} 40 & -32 & -1 \\ -26 & 25 & 0 \\ -15 & 6 & 8 \end{bmatrix}$

Ans: $\begin{bmatrix} -40 & 32 & 1 \\ 26 & -25 & 0 \\ 15 & -6 & -8 \end{bmatrix} \div \begin{bmatrix} 1 & 40 & -2 & 13 & -15 \\ 40 & -26 & -15 \end{bmatrix}$
 (-1)

Q5) (a)

Ans: Interchange $R_1 \leftrightarrow R_2$

(b) $\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -1 & 2 & -9 \end{bmatrix}$

$(-1)[-36] \quad (-1)[3+16]$

(a) $[-27+8] \quad (1)[-2+12]$

$(-1)[4-6]$

(2) $[-1+8]$

(4) $[-7-16]$

$\begin{bmatrix} 38 & -42 & -80 \\ -42 & -28 & 20 \\ -76 & 14 & 90 \end{bmatrix} \rightarrow \begin{bmatrix} 38 & -42 & -76 \\ 42 & -28 & -14 \\ -80 & -10 & 90 \end{bmatrix}$

$38+42-80$

$0/\text{NOT exists}$

$-1[-36-2] -3[-18+9]$

$-4[4+16]$

$38+42-80$

$0/\text{NOT exists}$

(a) $\begin{bmatrix} 1/5 & 1/5 & -2/5 \\ 2/5 & 2/5 & 3/20 \\ 2/5 & -1/5 & 1/10 \end{bmatrix}$

Ans: $\begin{bmatrix} 1/5 & 1/5 & -2/5 \\ 2/5 & 2/5 & 3/20 \\ 2/5 & -1/5 & 1/10 \end{bmatrix}$

$\frac{2}{5}[\frac{1}{5} - \frac{1}{5}] \quad \frac{2}{5}[\frac{1}{5} + \frac{2}{5}]$

$\frac{1}{5}[\frac{2}{5} - \frac{8}{25}] \quad \frac{1}{5}[\frac{6}{5} + \frac{25}{5}]$

$\frac{1}{5}[\frac{2}{5} - \frac{8}{25}] \quad \frac{1}{5}[\frac{6}{5} + \frac{25}{5}]$

$\begin{bmatrix} -3/250 & 0 & 6/25 \\ -13/250 & 0 & 7/25 \\ -17/250 & 0 & 1/25 \end{bmatrix}$

$\frac{1}{50}$

$\frac{1}{50}$

$\frac{1}{50}$

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$$\text{ii). } \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 2 & 5 & 3 & | & 0 & 1 & 0 \\ 1 & 0 & 8 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 1 & 0 & 8 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_1 + R_2$$

$$\begin{bmatrix} 1 & 3 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 1 & 0 & 8 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 3 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & -3 & 8 & | & 1 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \rightarrow R_3 + 3R_2$$

$$\begin{bmatrix} 1 & 3 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \rightarrow R_2 + 3R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & | & -10 & 16 & 9 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \rightarrow R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & | & -10 & 16 & 9 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & 1 & | & 5 & -2 & -1 \end{bmatrix}$$

→ Inverse Answer

$$\begin{bmatrix} -10 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

$$\text{b). } \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 2 & 4 & 1 & | & 0 & 1 & 0 \\ -4 & 2 & -9 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_3 + 4R_1$$

$$\begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 2 & 4 & 1 & | & 0 & 1 & 0 \\ 0 & -10 & 7 & | & 4 & 0 & 1 \end{bmatrix} \rightarrow R_1 + 4R_2$$

$$\begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 7 & 19 & 0 & | & 1 & 4 & 0 \\ 0 & -10 & 7 & | & -4 & 0 & 1 \end{bmatrix} \rightarrow R_2 + 7R_1$$

$$\begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 0 & 4 & 0 & | & 0 & 4 & 0 \\ 0 & -10 & 7 & | & -4 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 2 & 4 & 1 & | & 0 & 1 & 0 \\ -4 & 2 & -9 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_2 + R_1$$

$$= \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ -4 & 2 & -9 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_3 - 4R_2$$

$$= \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & -10 & 19 & | & -8 & -4 & 1 \end{bmatrix} \rightarrow R_3 + R_2$$

$$= \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & 0 & 12 & | & -6 & -3 & 1 \end{bmatrix} \rightarrow 3R_1 + R_3$$

$$= \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & 0 & 12 & | & -6 & -3 & 1 \end{bmatrix}$$

Q12.

$$\text{Q13). } \begin{bmatrix} 1 & 0 & 1 & | & 2 & 0 & 0 \\ 0 & 1 & 1 & | & 0 & 1 & 0 \\ 1 & 1 & 0 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 & 0 & 0 \\ -1 & 1 & 0 & | & -2 & 1 & 0 \\ 1 & 1 & 0 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 & 0 & 0 \\ 0 & 2 & 0 & | & -2 & 1 & 1 \\ 1 & 1 & 0 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 & 0 & 0 \\ 0 & 2 & 0 & | & -2 & 1 & 1 \\ 2 & 0 & 0 & | & 2 & -1 & 1 \end{bmatrix} \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 0 & 1 & | & 1 & 0 & 0 \\ 0 & 2 & 0 & | & -2 & 1 & 1 \\ 0 & 0 & -2 & | & -2 & -1 & 1 \end{bmatrix} \rightarrow R_1 + R_3$$

$$\begin{bmatrix} 2 & 0 & 0 & | & 1 & -1 & 1 \\ 0 & 2 & 0 & | & -2 & 1 & 1 \\ 0 & 0 & -2 & | & -2 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{bmatrix}$$

Inverse

Ans.

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Q17).

Ans:-
$$\begin{bmatrix} \sqrt{2} & 3\sqrt{2} & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{2} & 3\sqrt{2} & 0 & 1 & 0 & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow R_2 + 4R_1$$

$$\begin{bmatrix} \sqrt{2} & 3\sqrt{2} & 0 & 1 & 0 & 0 \\ 0 & 13\sqrt{2} & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \frac{1}{\sqrt{2}} \rightarrow 13R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 3 & 0 & 13\sqrt{2} & 0 & 0 & 1 & -3 & 0 \\ 0 & 13\sqrt{2} & 0 & 4 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \div 13\sqrt{2} \rightarrow \text{Ans:}$$

$$\begin{bmatrix} \frac{1}{13\sqrt{2}} & -\frac{3}{13\sqrt{2}} & 0 \\ \frac{4}{13\sqrt{2}} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q15).

Ans:-
$$\begin{bmatrix} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 2 & 7 & 7 & 0 & 0 & 1 \end{bmatrix} \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \rightarrow R_3 - R_1$$

$$\begin{bmatrix} -2 & -6 & 0 & -1 & -6 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \rightarrow R_1 + 6R_2$$

$$-\frac{1}{2} \left[\begin{bmatrix} -2 & 0 & 0 & -7 & 0 & 6 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \right] \rightarrow \text{Ans:}$$

$$\begin{bmatrix} 7/2 & 0 & -3 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -4 & -32 & 6 & 0 & 0 \end{bmatrix} \rightarrow R_1 + R_2$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 24 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -4 & -32 & 6 & 0 & 0 \end{bmatrix} \rightarrow R_2 - 2R_3$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 24 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -4 & -32 & 6 & 0 & 0 \end{bmatrix} \rightarrow \frac{1}{2} \rightarrow \frac{1}{4}$$

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Q16).

Ans:-
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 3 & 5 & 0 & 0 & 0 & 1 & 0 \\ 1 & 3 & 5 & 7 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_4 - R_3 \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 3 & 5 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 7 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 - R_1 \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 7 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 7 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Ans:-
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1/3 & 4/3 & 1/3 \\ 0 & -1/5 & 1/5 & 1/5 \\ 0 & 0 & -1/7 & 1/7 \end{bmatrix}$$

Q17).

Ans:-
$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow 2R_2 - R_1$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 24 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow 4R_3 - R_1$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 24 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ -2 & 6 & -16 & -20 & -1 & 0 & 0 & 4 \end{bmatrix} \rightarrow R_1 + R_4$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 24 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -4 & -16 & -20 & -1 & 0 & 0 & 4 \end{bmatrix} \rightarrow 2R_4 + R_2$$

$$\begin{bmatrix} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 8 & 24 & 0 & -12 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -8 & -40 & -3 & 2 & 2 & 0 \end{bmatrix} \rightarrow 4R_3 + R_4$$